

II-Probabilistic Approaches Independence and Graphical Models

Foundations of Neuro-Symbolic AI

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Outline

- ▶ Independence of variables
- ▶ Conditional independence
- ▶ Decompositions given independence
- ▶ Bayesian networks
- ▶ Markov networks

How to beat the curse of dimensions?

Probability distribution of factored systems with states $\times_{k \in [d]} [m_k]$ has

$$\left(\prod_{k \in [d]} m_k \right) - 1$$

degrees of freedom (coordinates to specify and store).

Mitigation: [Tensor network decompositions](#)

Idea

Exploit the "dependence" structure of variables to find decompositions of probability distributions.

Example: At the dentist

Consider the joint distribution of the variables X_0 (Toothache), X_1 (Cavity), X_2 (Catch), with a variable X_3 , denoting whether there is a Cloud outside the dentists lab.

- ▶ We now have an order-4 probability tensor $\mathbb{P}[X_0, X_1, X_2, X_3]$
- ▶ Intuitively, knowing Cloud should not affect the probability of having a cavity, so why shall we care?

To be more precise let us consider the distribution:

$$\mathbb{P}[X_0, X_1, X_2, X_3] = \begin{pmatrix} 0.288 & 0.004 \\ 0.032 & 0.006 \end{pmatrix} \begin{pmatrix} 0.077 & 0.036 \\ 0.008 & 0.054 \end{pmatrix} [X_0, X_1, X_2, X_3]$$

Such redundancies are captured by the concept of independence.

Formal definition of Independencies

Definition (Independence)

Given a joint distribution $\mathbb{P}[X_{\mathcal{V}}]$ and two disjoint subsets $A, B \subset \mathcal{V}$, we say that the variables X_A are independent from the variables X_B if the marginal distributions $\mathbb{P}[X_{A \cup B}]$, $\mathbb{P}[X_A]$ and $\mathbb{P}[X_B]$ satisfy for any index tuples x_A, x_B

$$\mathbb{P}[X_A = x_A, X_B = x_B] = \mathbb{P}[X_A = x_A] \cdot \mathbb{P}[X_B = x_B] .$$

We denote in this case $(X_A \perp X_B)$

Interpretation: Knowing the value of variables B (and conditioning on them) does not change the distribution of variables A .

Contraction criterion for independence

Theorem

If and only if the X_A are independent from the variables X_B with respect to a distribution $\mathbb{P}[X_V]$ then

$$\langle \mathbb{P}[X_{[d]}] \rangle_{[X_A, X_B]} = \langle \mathbb{P}[X_{[d]}] \rangle_{[X_A]} \otimes \langle \mathbb{P}[X_{[d]}] \rangle_{[X_B]} .$$

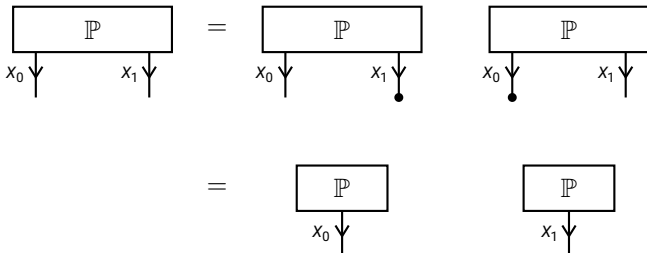
In a tensor network diagram this is the decomposition:

The diagram illustrates the decomposition of a joint distribution tensor into a tensor product of marginal distributions. On the left, a single box labeled $\langle \mathbb{P}[X_{[d]}] \rangle_{[X_A, X_B]}$ has four vertical legs extending downwards. The first two legs are grouped by a brace and labeled X_A , and the last two legs are grouped by a brace and labeled X_B . This is followed by an equals sign. On the right, there are two separate boxes. The first box is labeled $\langle \mathbb{P}[X_{[d]}] \rangle_{[X_A]}$ and has two vertical legs labeled X_A . The second box is labeled $\langle \mathbb{P}[X_{[d]}] \rangle_{[X_B]}$ and has two vertical legs labeled X_B . The entire equation is represented as:

$$\langle \mathbb{P}[X_{[d]}] \rangle_{[X_A, X_B]} = \langle \mathbb{P}[X_{[d]}] \rangle_{[X_A]} \otimes \langle \mathbb{P}[X_{[d]}] \rangle_{[X_B]}$$

Storage of distributions by marginal distributions

Independence allows the decomposition into tensor products



Exponential to linear storage demand

Instead of storing $m_0 \cdot m_1$ coordinates, we can store $\mathbb{P}$ in the case of independency with $m_0 + m_1$ demand.

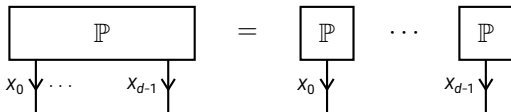
Decomposition of pairwise independent variables

Imagine now that we have a distribution $\mathbb{P} [X_{[d]}]$, such that for any $k, \tilde{k} \in [d]$

$$(X_k \perp X_{\tilde{k}})$$

We then have the decomposition:

$$\mathbb{P} [X_{[d]}] = \bigotimes_{k \in [d]} \mathbb{P} [X_k]$$



Exponential to linear storage demand

Instead of storing $\prod_{k \in [d]} m_k$ coordinates, we can store $\mathbb{P}$ using $\sum_{k \in [d]} m_k$ coordinates of the marginal distributions.

Outline

- ▶ Independence of variables
- ▶ **Conditional independence**
- ▶ Decompositions given independence
- ▶ Bayesian networks
- ▶ Markov networks

Assumption of independence

Independence is a very restrictive assumption:

- ▶ We need dependence of effects on causes for machine learning to work.
- ▶ Independence covers only the most extreme form of tensor network decompositions.

Example: In the dentist example (without the **Cloud** variable), no variables are independent from each other.

Are there more common properties corresponding to more generic tensor decompositions?

Formal definition of Conditional Independencies

Definition (Conditional Independence)

Given a joint distribution of variables X_A , X_B and X_C , we say that X_A and X_B are independent given X_C if for any index tuple (x_A, x_B, x_C)

$$\mathbb{P}[X_A = x_A, X_B = x_B | X_C = x_C] = \mathbb{P}[X_A = x_A | X_C = x_C] \cdot \mathbb{P}[X_B = x_B | X_C = x_C],$$

We denote this situation by

$$(X_A \perp X_B) \mid X_C.$$

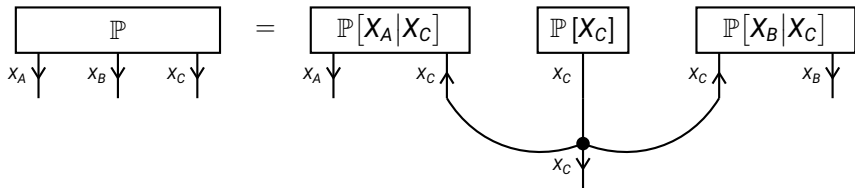
Interpretation: Conditional independence means that X_A and X_B are independent with respect to any slice $\mathbb{P}[X_A, X_B | X_C = x_C]$.

Conditional independence as a contraction equation

We have $(X_A \perp X_B) \mid X_C$ if and only if

$$\mathbb{P}[X_A, X_B \mid X_C] = \langle \mathbb{P}[X_A \mid X_C], \mathbb{P}[X_B \mid X_C] \rangle_{[X_A, X_B, X_C]} .$$

We depict conditional independence by tensor network decompositions:



Example: Being at a dentist

Task: Let us investigate, whether **Toothache** (X_0) and **Catch** (X_2) are independent given **Cavity** (X_1), with respect to the distribution:

$$\mathbb{P}[X_0, X_1, X_2] = \begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix} \begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix} [X_0, X_1, X_2]$$

For $X_1 = 0$:

For $X_1 = 1$:

Example: Being at a dentist

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For $X_1 = 0$:

$$\mathbb{P}[X_0, X_2 | X_1 = 0] = \begin{pmatrix} 0.72 & 0.08 \\ 0.18 & 0.02 \end{pmatrix} [X_0, X_2]$$

For $X_1 = 1$:

Example: Being at a dentist

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For $X_1 = 0$: **Independent** ✓

$$\mathbb{P}[X_0, X_2 | X_1 = 0] = \begin{pmatrix} 0.72 & 0.08 \\ 0.18 & 0.02 \end{pmatrix} [X_0, X_2] = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} [X_0] \otimes \begin{pmatrix} 0.9 \\ 0.1 \end{pmatrix} [X_2]$$

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For $X_1 = 1$: **Independent** ✓

$$\mathbb{P}[X_0, X_2 | X_1 = 1] = \begin{pmatrix} 0.04 & 0.36 \\ 0.06 & 0.54 \end{pmatrix} [X_0, X_2]$$

Example: Being at a dentist

Task: Let us investigate, whether **Toothache** (X_0) and **Catch** (X_2) are independent given **Cavity** (X_1), with respect to the distribution:

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For $X_1 = 1$: **Independent** ✓

$$\mathbb{P}[X_0, X_2 | X_1 = 1] = \begin{pmatrix} 0.04 & 0.36 \\ 0.06 & 0.54 \end{pmatrix} [X_0, X_2] = \begin{pmatrix} 0.4 \\ 0.6 \end{pmatrix} [X_0] \otimes \begin{pmatrix} 0.1 \\ 0.9 \end{pmatrix} [X_2]$$

Example: Being at a dentist

Task: Let us investigate, whether **Toothache** (X_0) and **Catch** (X_2) are independent given **Cavity** (X_1), with respect to the distribution:

$$\mathbb{P}[X_0, X_1, X_2] = \begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix} \begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix} [X_0, X_1, X_2]$$

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Conclusion: The conditional independence holds, that is $(X_0 \perp X_2) | X_1$.

Outline

- ▶ Independence of variables
- ▶ Conditional independence
- ▶ **Decompositions given independence**
- ▶ Bayesian networks
- ▶ Markov networks

Chain Rule: Decomposing Probabilities

Theorem (Chain Rule)

For any joint probability distribution $\mathbb{P}[X_{[d]}]$ we have

$$\mathbb{P}[X_{[d]}] = \langle \{ \mathbb{P}[X_k | X_{[k]}] : k \in [d] \} \rangle_{[X_{[d]}]} .$$

Proof sketch: We apply the Bayes theorem to get

$$\mathbb{P}[X_{[d]}] = \langle \mathbb{P}[X_0], \mathbb{P}[X_1, \dots, X_{d-1} | X_0] \rangle_{[X_{[d]}]} .$$

and further apply the Bayes theorem iteratively on the conditioned distribution.

Problem: The tensors $\mathbb{P}[X_k | X_{[k]}]$ are of order $d + 1$!

Can we smallen the order based on independence assumptions?

Simplification of conditional distributions

Theorem

When $(X_A \perp X_B) \mid X_C$ then

$$\mathbb{P}[X_A|X_C, X_B] = \mathbb{P}[X_A|X_C] \otimes \mathbb{I}[X_B] .$$

Proof sketch: We have by Bayes theorem

$$\mathbb{P}[X_A, X_B|X_C] = \langle \mathbb{P}[X_A|X_C, X_B], \mathbb{P}[X_B|X_C] \rangle_{[X_A, X_B, X_C]}$$

and compare with the conditional independence equation

$$\mathbb{P}[X_A, X_B|X_C] = \langle \mathbb{P}[X_A|X_C], \mathbb{P}[X_B|X_C] \rangle_{[X_A, X_B, X_C]} .$$

Usage for decompositions: We can drop some conditioned variables and effectively smallen the tensor order in the tensors $\mathbb{P}[X_k|X_{[k]}]$.

Example: Markov Chains

Let us model the weather in a town on all days of a year by variables $X_{[365]}$ and assume that for each $k \in [365]$

- ▶ Three states $x_k = 0$ (sunny), $x_k = 1$ (cloudy), $x_k = 2$ (rainy)
- ▶ The weather on the day $k + 1$ conditioned on the weather on the day k is

$$\mathbb{P}[X_{k+1}|X_k] = \begin{pmatrix} 0.70 & 0.25 & 0.05 \\ 0.35 & 0.40 & 0.25 \\ 0.10 & 0.40 & 0.50 \end{pmatrix} [X_{k+1}, X_k]$$

- ▶ The weather on the day $k + 1$ does not depend on the weather before day k given the weather before day k :

$$(X_{k+1} \perp X_{[k]}) \mid X_k$$

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- ▶ The weather on the day $k + 1$ does not depend on the weather before day k given the weather before day k :

$$(X_{k+1} \perp X_{[k]}) \mid X_k$$

This is an example of a Markov chain: Variables depend only on their predecessors.

Example: Markov Chains

Theorem (Markov Chain)

Let there be a set of variables X_k where $k \in [d]$. When X_k is independent of $X_{[k-1]}$ given X_{k-1} (the Markov Property), then

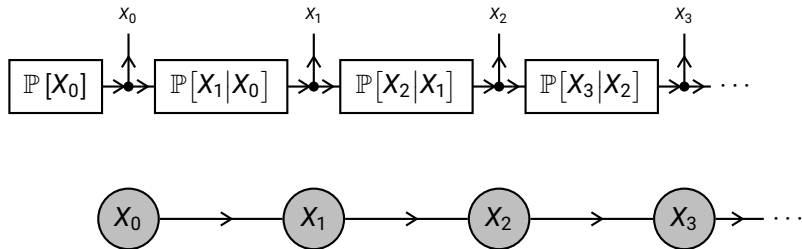
$$\mathbb{P}[X_{[d]}] = \langle \mathbb{P}[X_k | X_{k-1}] : k \in [d] \rangle_{[X_{[d]}]} .$$

Proof sketch: We use the chain decomposition and exploit the independence assumptions to get

$$\mathbb{P}[X_k | X_{[k]}] = \mathbb{P}[X_k | X_{k-1}] \otimes \mathbb{I}[X_{[k-1]}] .$$

Tensor network decompositions of markov chains

We depict this by



Can we find a more generic correspondence between probability decompositions and conditional independence statements?

Outline

- ▶ Independence of variables
- ▶ Conditional independence
- ▶ Decompositions given independence
- ▶ **Bayesian networks**
- ▶ Markov networks

Independence assumptions of Bayesian networks

Towards decomposing a distribution in a more generic way take the steps:

- ▶ We choose an order and enumerate the variables by $[d]$.
- ▶ To each $k \in [d]$ we choose a set $\text{Pa}(k)$ of **parent nodes** in $[k] = \{0, \dots, k-1\}$ such that

$$(X_k \perp X_{[k] \setminus \text{Pa}(k)}) \mid X_{\text{Pa}(k)}$$

Example: Markov chains are the special case of $\text{Pa}(k) = \{k-1\}$.

Factorization into Bayesian networks

Theorem

When for each $k \in [d]$

$$(X_k \perp X_{[k] \setminus \text{Pa}(k)}) \mid X_{\text{Pa}(k)}$$

then

$$\mathbb{P}[X_{[d]}] = \langle \{\mathbb{P}[X_k | X_{\text{Pa}(k)}] : k \in [d]\} \rangle_{[X_{[d]}]} .$$

Proof sketch: As in the proof of the Markov chain factorization theorem, we use the chain factorization and exploit the conditional independences to show that

$$\mathbb{P}[X_k | X_{[k]}] = \mathbb{P}[X_k | X_{\text{Pa}(k)}] \otimes \mathbb{I}[X_{[k] \setminus \text{Pa}(k)}] .$$

Example: At the Dentist

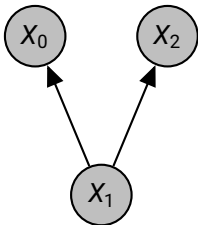
- Choose the order X_1, X_0, X_2
- For both X_0 and X_2 we take X_1 as single parent, that is

$$\text{Pa}(0) = \{1\} \quad \text{and} \quad \text{Pa}(2) = \{1\}$$

and have the for the Bayesian network necessary conditional independencies (shown before)

$$(X_0 \perp X_2) \mid X_1 \quad \text{and} \quad (X_2 \perp X_0) \mid X_1$$

The Toothache



$$\mathbb{P}[X_0, X_1, X_2] = \mathbb{P}[X_0|X_1] \mathbb{P}[X_2|X_1] \mathbb{P}[X_1]$$

The equation is accompanied by a plate notation diagram. It shows two rectangular plates side-by-side. The left plate contains the expression $\mathbb{P}[X_0|X_1]$ and has an upward arrow labeled X_0 pointing from its top. The right plate contains the expression $\mathbb{P}[X_2|X_1]$ and has an upward arrow labeled X_2 pointing from its top. Below these two plates, there is a single upward arrow labeled X_1 pointing to a small black dot. From this dot, two curved arrows point upwards, one to the bottom of the left plate and one to the bottom of the right plate, indicating that the variable X_1 is shared as a parent for both X_0 and X_2 .

Example: Hidden Markov Models

Extend the weather example: We observe the weather E_k , but not the hidden driving variable:

- Atmospheric pressure X_k taking states $x_k = 0$ (high pressure) or $x_k = 1$ (low pressure)

Transition of the pressure variable:

$$\begin{pmatrix} 0.80 & 0.20 \\ 0.30 & 0.70 \end{pmatrix} [X_{k+1}, X_k]$$

Probability of whether given the

$$\begin{pmatrix} 0.70 & 0.25 & 0.05 \\ 0.10 & 0.40 & 0.50 \end{pmatrix} [E_k, X_k]$$

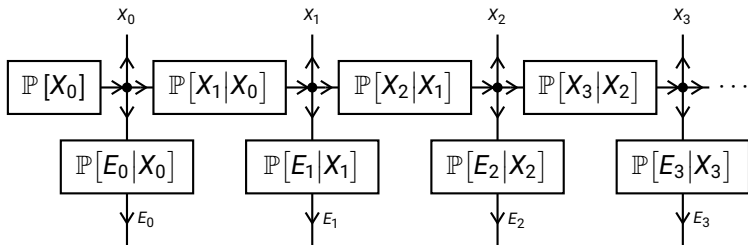
This is an example of Hidden Markov Models extending Markov Chains by limited observation E_k of the variables X_k .

Example: Hidden Markov Models

The independence assumptions of a Hidden Markov Model are for any $k \in [d]$

- ▶ X_{k+1} is independent of $X_{[k]}$ and $E_{[k]}$ given X_k
- ▶ E_k is independent of all other variables given X_k

The independence assumptions are exploited in the decomposition



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Undirected Graphical Models: Markov Networks

Definition (Markov Networks)

Let $\{\tau^e[X_e] : e \in \mathcal{E}\}$ be a Tensor Network on a hypergraph $\mathcal{G}$. The associated Markov Network is the probability distribution of $\{X_v : v \in \mathcal{V}\}$ defined by

$$\mathbb{P}[X_{\mathcal{V}}] = \frac{\langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{V}}]}}{\langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[\emptyset]}} = \langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{V}}|\emptyset]} .$$

We call the denominator

$$\mathcal{Z}(\mathcal{G}) = \langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[\emptyset]}$$

the **partition function** of the Markov Network.

Markov network independencies

Theorem

A positive distribution $\mathbb{P}[X_{\mathcal{V}}]$ has a representation as a Markov network on a hypergraph $\mathcal{G}$, if and only if for any disjoint subsets $\mathcal{U}, \mathcal{W}, \mathcal{I} \subset \mathcal{V}$ for which $\mathcal{I}$ separates $\mathcal{U}$ from $\mathcal{W}$

$$(X_{\mathcal{U}} \perp X_{\mathcal{W}}) \mid X_{\mathcal{I}}.$$

Proof sketch: This is the Hammersley Clifford factorization theorem, which proof is out of scope in this course.